



Q4

Part a

General Overview

The logic for the serial code (found in `src/src_serial`) is as follows

- Load in the config file (`config/config.txt`) from the user
- Initialise all variables necessary
- Start a `while (true)` loop
- Within the loop, circle centers are generated via a seeded random number generator. Those circles are checked against an `unordered_map` of the other generated circles for overlap.
- If no overlap is found, it is placed within the `unordered_map`
- After the user parameter `sampling_frequency` number of trials, there is a check for convergence. If the number of circles generated has not changed after `sampling_frequency` trials, then there can be no more circles placed without overlap with these parameters and the loop is exited.
- The final results are outputted and graphs can be plotted (`graphing/graph.py`).

The use of `make` with the rules `serial` , `serial_debug` , `serial_vis` are used to compile the source code. Each rule gives a different binary output with:

- `serial` : Runs the binary without any visualisation or debugging, and outputs final answer.
- `serial_debug` : Runs the binary with custom debugging flags.
- `serial_vis` : Runs the binary with custom visualisation flags, that continually outputs the key data for visualisation purposes (you can watch the numbers go up).

I highly recommend using the bash script `run_serial.sh` with command `bash ./run_serial <MAKE_RULE>` . This builds and executes the binary, and will create

directories for results to be moved into, which then can be plotted using the Python script.

Specfics

Generation of circles

```
/* Generates and returns a Point, within the boundaries */
Point gen_random_pair(rng &random_gen)
{
    return {
        r + (L - 2 * r) * random_gen.grnd(),
        r + (L - 2 * r) * random_gen.grnd(),
    };
}
```

The generating function will generate a random number taht has been rpreviously seeded, with a number that is within the boundaries of the box. This assures that no circle can overlap the boundary, and reduces the need to run four comparison checks every generation.

Checking overlap

The biggest bottleneck to this problem is the checking the overlap of circles. A naive approach would be a simple comparison between every circle, giving an $O(N^2)$ loop. For efficency, I implemented a hash map that relates `Cells` to `vector<Point>` i.e I split the grid up into cells that are determined by the input parameters `L` and `r` i.e. `grid_size = static_cast<int>(L / (2 * r));`.

```

// Check if a new circle overlaps with circles in nearby grid cells
bool check_overlap(const Point &new_circle)
{
    Cell cell = get_grid_cell(new_circle);

    const vector<Cell> neighbours = {{-1, -1}, {-1, 0}, {-1, 1},
    {0, -1}, {0, 0}, {0, 1}, {1, -1}, {1, 0}, {1, 1}};

    // Check nearby neighbour cells

    for (auto &neighbour : neighbours)
    {
        Cell neighbor_cell = {cell.first + neighbour.first,
        cell.second + neighbour.second};

        // Check if neighbor cell exists in the spatial grid
        if (grid.find(neighbor_cell) != grid.end())
        {
            for (const auto &existing_circle : grid[neighbor_cell])
            {
                // Compute squared distance for comparison
                double dx = existing_circle.first - new_circle.first;
                double dy = existing_circle.second - new_circle.second;
                double distance_squared = dx * dx + dy * dy;

                if (distance_squared < r_comp)
                {
                    return true; // If overlap, return true
                }
            }
        }
    }

    return false; // No overlap found
}

```

The `check_overlap` function will only compare the circles that are within the nearest neighbour cells. This is a lot more efficient than $O(N^2)$, as a hash map

(with a good hash) is on average $O(1)$ lookup time.

Sampling Frequency approach

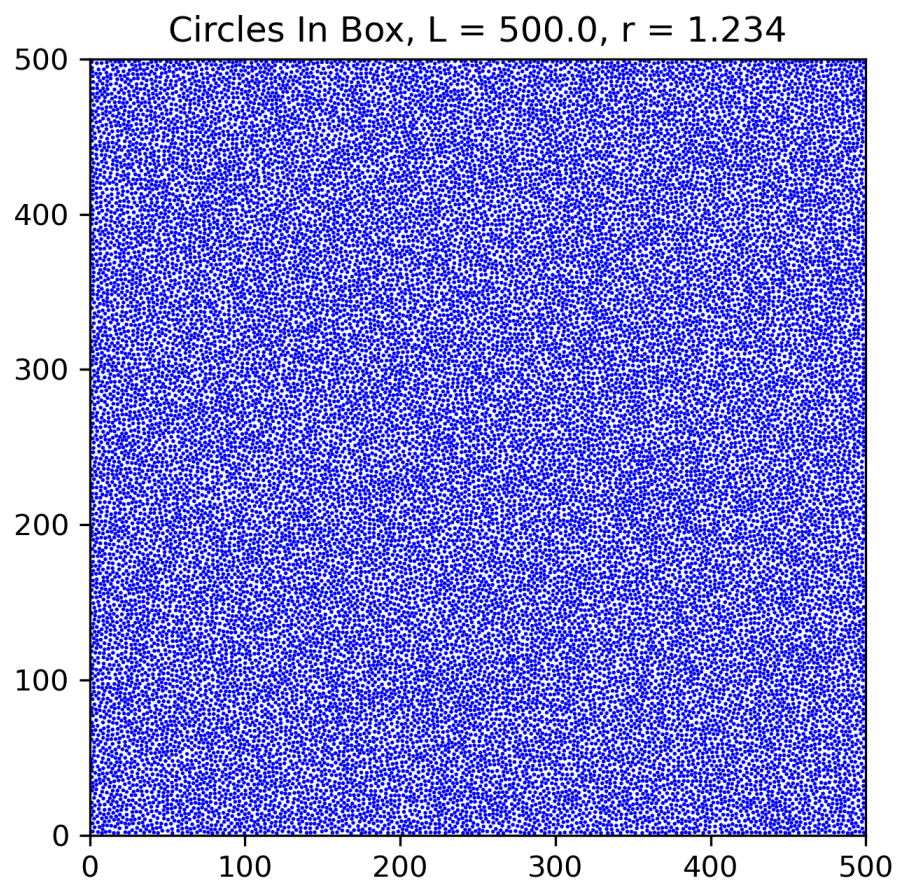
I implemented a `sampling_frequency` approach to the generation of circles, to fit into the idea of "run until no more circles can be placed". Choosing a small `sampling_frequency` will be more efficient, however is very likely to miss a lot of circles that could be generated. On the other hand, choosing a large `sampling_frequency` will be less efficient, but you will likely catch a lot of circles that could be generated. An improvement that could be made is to make the parameter dynamic. I found that a parameter of `1e6` is a good benchmark, however results have been given for `1e6` , `1.25e6` , `1.50e6` , and `1e7` . These can also be found in `results/serial_results/L=500, r=1.234/sf={SAMPLING_FREQUENCY}` . All these results were run on `VIKING` with `job_serial.sh` .

Results

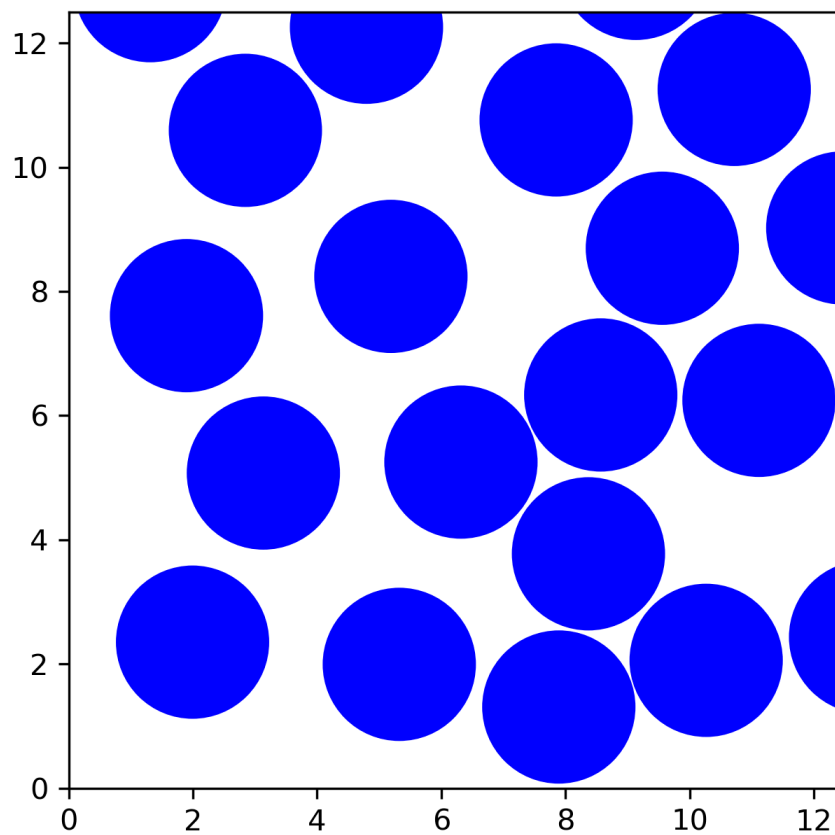
Table

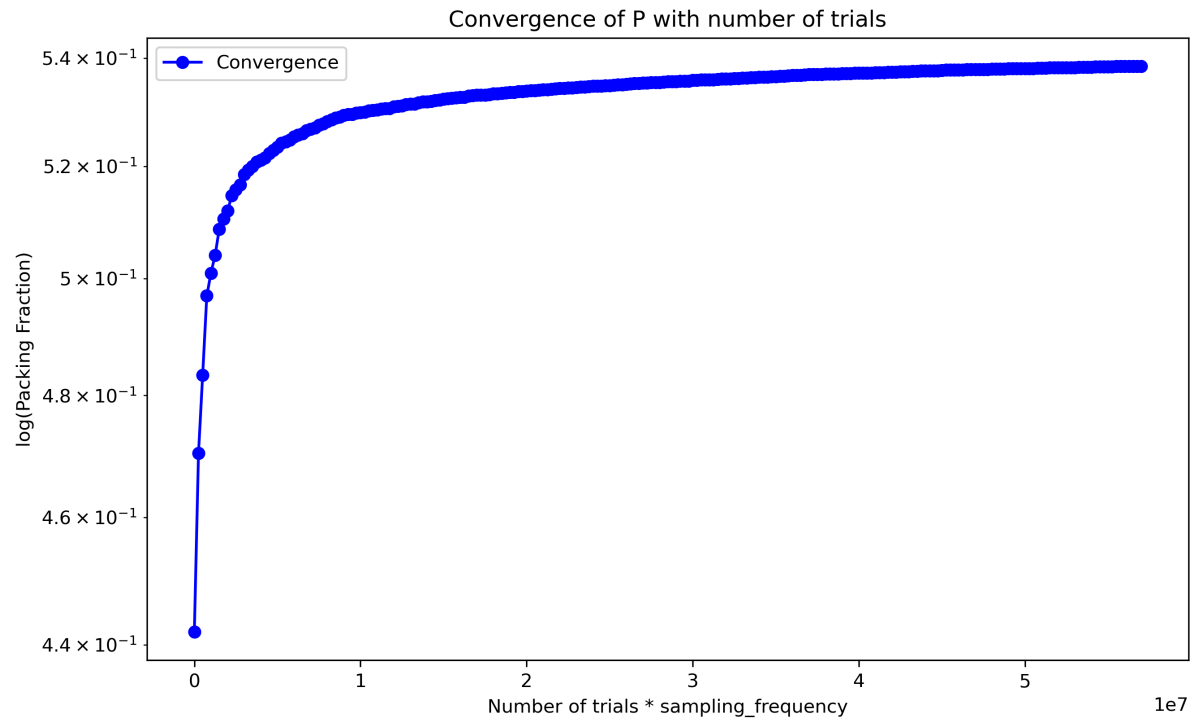
Sampling Frequency	Number of Circles Placed	Packing Fraction	Time(s)
1e6	28141	0.538493	176.691515
1.25e6	28199	0.539602	253.402379
1.50e6	28226	0.540119	290.254791
1e7	28335	0.542205	841.623180

Plots



Circles in Smaller Box, $L = 500.0$, $r = 1.234$





The above plots are for `sampling frequency=1e6`. The plots across all sampling frequencies are very similar, and more can be found in `./results/serial_results`. Note for the convergence plot, at every `sampling_frequency / 4` intervals and after each generation of trials, a point is plotted. This is why the x-axis is labelled `Number of trials * sampling_frequency`.

Part b

General Overview

The logic for the parallel code (found in `src/src_parallel`) is as follows:

The logic for the serial code (found in `src/src_serial`) is as follows

- Load in the config file (`config/config.txt`) from the user
- Initialise all variables necessary
- Initialise an OpenMP parallel region
- Initialise a thread-local random number generator with a unique seed depending on the thread id and total number of threads.

- Start a `while` loop
- Within the loop, circle centers are generated in parallel, and added to a new vector `trial_placements` with size `sampling_frequency`. Each circle has a flag associated with it.
- Trials are checked in parallel with the grid of placed circles for overlap. If an overlap is found, the trial's flag is set.
- Trials are checked in parallel within themselves, skipping any trials that have flags that are already set.
- Trials that have not got a set flag are placed in the grid.
- There is a check for convergence. If the number of circles generated has not changed after `sampling_frequency` trials, then there can be no more circles placed without overlap with these parameters and the loop is exited.

The use of `make` with the rules `parallel`, `parallel_debug`, `parallel_vis` are used to compile the source code. Each rule gives a different binary output with:

- `parallel` : Runs the binary without any visualisation or debugging, and outputs final answer.
- `parallel_debug` : Runs the binary with custom debugging flags.
- `parallel_vis` : Runs the binary with custom visualisation flags, that continually outputs the key data for visualisation purposes (you can watch the numbers go up).

I highly recommend using the bash script `run_parallel.sh` with command `bash ./run_parallel <MAKE_RULE> <NUM_OF_THREADS>`. This builds and executes the binary, and will create directories for results to be moved into, which then can be plotted using the Python script (note: The packing fraction convergence is not plotted here).

Why OpenMP

I chose to use OpenMP due to the clear cut logic that is required. Generating random points is a task that can easily be done in parallel, same with read only checking. OpenMP provides easy to use synchronisation in the form of barriers and critical statements, so updating shared resources like the grid is already controlled. OpenMP excels for dynamic workloads, utilising all threads effectively.

I did not want to use MPI for splitting up the grid, as for example, if spilt the grid into chunks and had each chunk generate circles, there will likely be correlation across the boundaries of the chunks, leading to space between the boundaries that other circles could be placed.

Scaling Tests

Thread Scaling

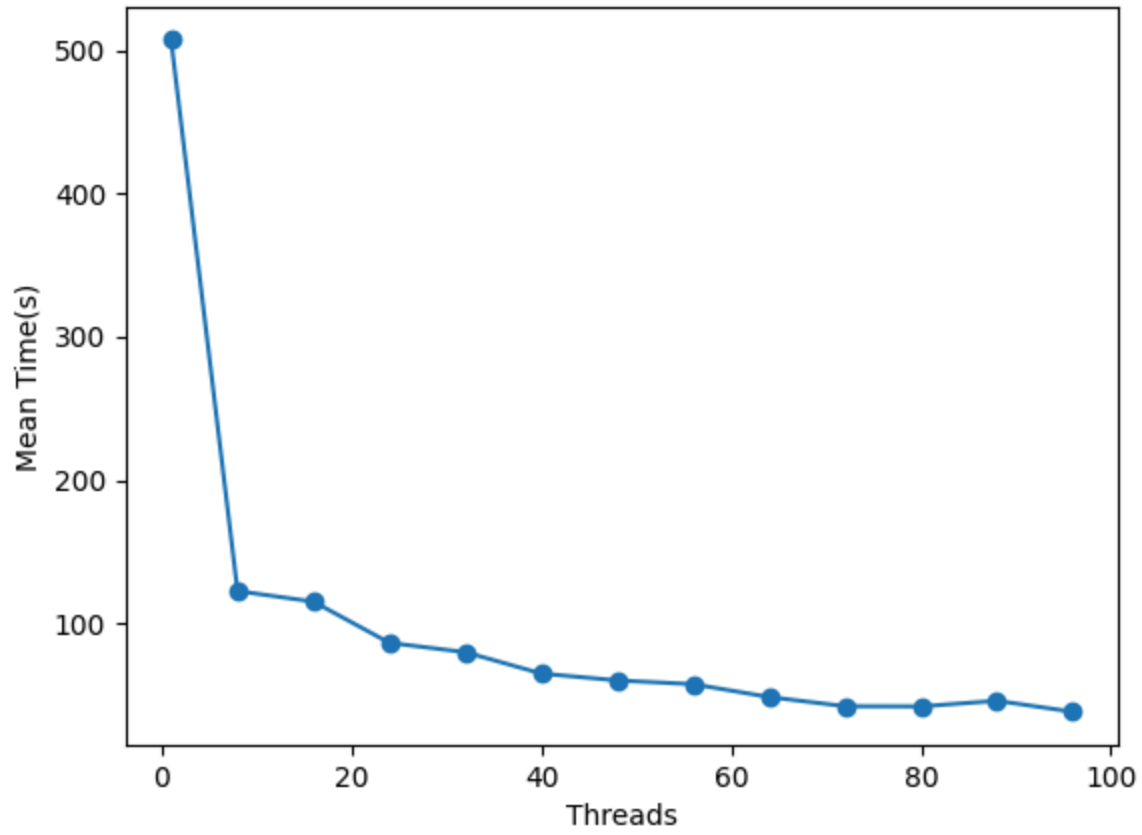
DESCRIBE PROBLEM HERE

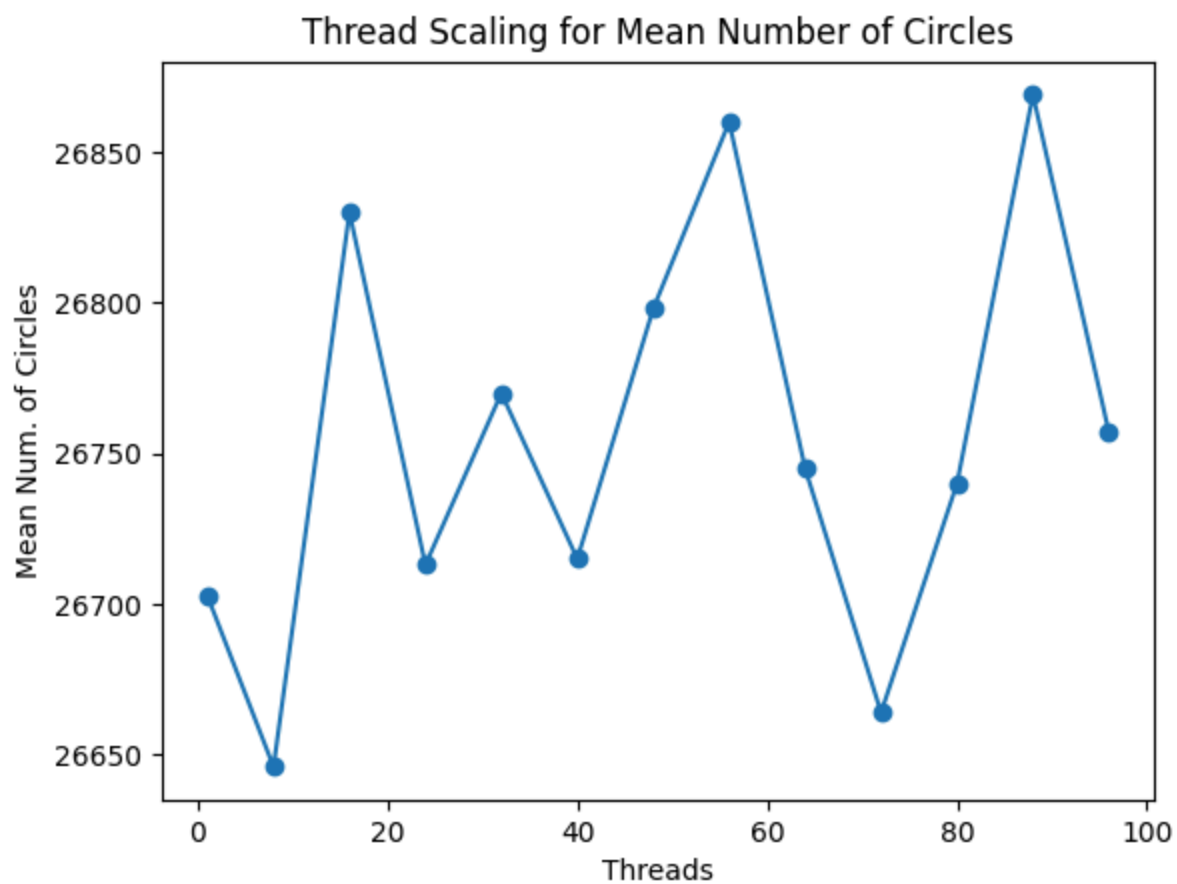
Number of Threads	Mean Time (s)
1	507.266506
8	122.714003
16	115.333433
24	86.598033
32	80.061899
40	65.159961
48	60.394195
56	57.760489
64	48.780394
72	42.371885
80	42.234132
88	46.167095
96	38.491898

Number of Threads	Mean Number of Circles
1	26703

Number of Threads	Mean Number of Circles
8	26646
16	26830
24	26713
32	26770
40	26715
48	26798
56	26860
64	26745
72	26664
80	26740
88	26869
96	26757

Thread Scaling for Mean Time





r / L ratios

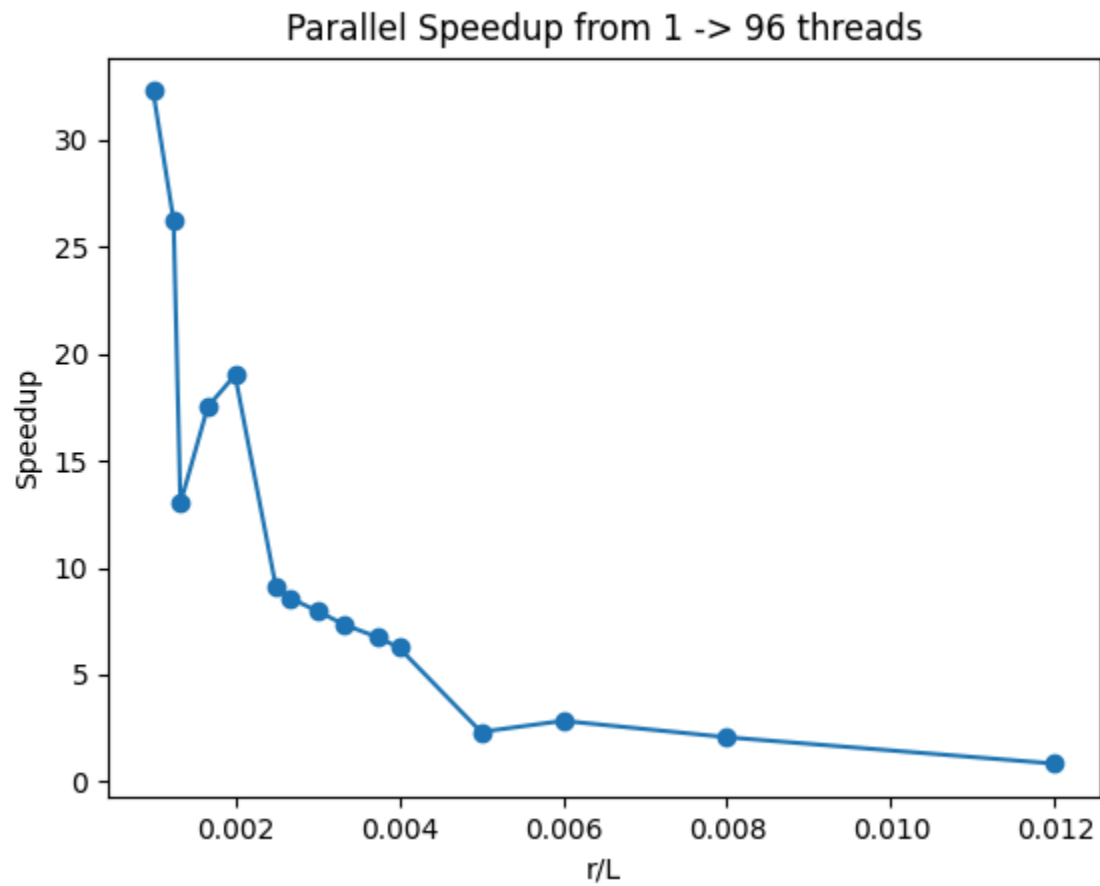
DESCRIBE PROBLEM HERE

r/L Ratio	Time (1 Thread) (s)	Time (96 Threads) (s)	Speedup (1 Thread / 96 Threads)
1/ 1000	610.832874	18.910452	32.30
1/800	299.347380	11.403005	26.24
1/750	142.934938	10.990958	13.01
1/600	134.451639	7.671533	17.52
2/ 1000	105.762529	5.563378	19.01

r/L Ratio	Time (1 Thread) (s)	Time (96 Threads) (s)	Speedup (1 Thread / 96 Threads)
2/800	42.947283	4.737450	9.07
2/750	31.434714	3.662494	8.58
3/ 1000	24.935044	3.125215	7.98
2/600	21.236937	2.895760	7.33
3/800	18.573556	2.759496	6.73
3/750	17.048064	2.725282	6.26
3/600	4.771479	2.057621	2.32
3/500	3.767288	1.328654	2.84
2/250	2.240967	1.083615	2.07
3/250	0.569546	0.684059	0.83

r/L Ratio	Num of Circles (1 Thread)	Num of Circles (96 Threads)	Difference (96 - 1)
1/1000	166986	166815	-171
1/800	106641	106327	-314
1/750	92355	93457	1102
1/600	59432	59455	23
2/1000	41159	41029	-130
2/800	25960	26097	137
2/750	22625	22840	215
3/1000	17869	17953	84
2/600	14409	14457	48

r/L Ratio	Num of Circles (1 Thread)	Num of Circles (96 Threads)	Difference (96 - 1)
3/800	11366	11449	83
3/750	9971	9795	-176
3/600	6223	6328	105
3/500	4301	4275	-26
2/250	2393	2385	-8
3/250	980	1024	44



Num. of Circle difference from 1 -> 96 threads

